OFFICIAL USER MANUAL & OPERATING GUIDELINES

# ErrinILS

## School Integrated Library System

Installation, daily circulation desk, automated 5-source cataloguing, student management, collection weeding, QR remote device pairing, analytics, and administrative recovery.



### Version 1.3.1 (Windows Edition)

Zero-Dependency Architecture • Standalone C# Executable

| | | | |
|---|---|---|---|
| **Target OS:** | Windows 7, 8, 8.1, 10, 11 (32/64-bit) | **Runtime:** | Microsoft .NET Framework 4.0+ |
| **Database:** | Single JSON File (`errin-data.json`) | **Release Date:** | November 2026 |
| **License:** | MIT Open Source License | **Web Portal:** | https://rzsy.github.io/ErrinILS/ |

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### Quick Role Guidance

**Librarians & Administrators:** Read all chapters. Start with Chapters 3 to 5, then proceed to Chapters 6 through 16.
**Student Helpers & Prefects:** Focus on Chapter 4.4 (scanners), Chapter 7 (Circulation Desk), and Chapter 8 (Simplified Kiosk).
**IT Technicians:** Focus on Chapters 2, 3, 13 (Remote QR), 15 (Backups), and 18 (Troubleshooting).

# 1. About ErrinILS

## 1.1 What ErrinILS is

**ErrinILS** is an integrated library system (ILS) designed specifically for school libraries. It is a lightweight Windows desktop program that maintains synchronized, audit-proof records of three operational elements:

- The **books and learning items** in the library collection.
- The **students, teachers, and patrons** registered to borrow them.
- The **active loans, historical borrowings, holds (reservations), and fines** connecting patrons and items.

Everyday library operation does not need an internet connection. All information is kept on the library computer and is saved automatically each time you make a change. The internet is used only for two optional tasks: looking up book bibliographic details by ISBN, and searching remote library catalogues via Z39.50.

## 1.2 Core Philosophy & Zero-Dependency Architecture

Traditional library systems (such as Koha or SLiMS) require heavy server stacks: Apache, PHP, MySQL, and Linux administration. ErrinILS runs on a **zero-external-dependency rule**. Built with pure C# on base .NET Framework 4.0 assemblies (`mscorlib`, `System`, `System.Drawing`, `System.Windows.Forms`), it compiles into a standalone ~200 KB executable (`ErrinILS.exe`) storing all data in human-readable JSON (`%APPDATA%\Errin\errin-data.json`).



# 2. System Requirements & Hardware Guidelines

ErrinILS is deliberately light, running smoothly on older classroom PCs without servers or database daemons.

| Component | Minimum Requirement | Recommended Specification |
|---|---|---|
| **Operating System** | Windows 7 SP1 (32 or 64-bit) | Windows 10 or Windows 11 (64-bit) |
| **Runtime** | Microsoft .NET Framework 4.0 | Microsoft .NET Framework 4.8 (built-in on Win 10/11) |
| **Memory (RAM)** | 1 GB RAM | 2 GB RAM or more |
| **Disk Space** | 10 MB for program; ~50 MB for data | 100 MB free storage |
| **Display Screen** | 1024 × 600 (min: 900 × 600) | 1366 × 768 or 1920 × 1080 (High-DPI PerMonitorV2) |
| **Scanner** | Keyboard wedge scanner (USB) | Standard USB 1D laser or 2D image scanner |
| **Printer** | Optional | A4 printer or thermal label printer (Code 128 barcodes) |



# 3. Installing, Updating and Starting ErrinILS

### 3.1 Installing with Setup program

Download `ErrinILS-Setup-1.3.1.exe`. Double-click it, click **Yes** at the UAC prompt, and if Windows SmartScreen appears, click **More info** then **Run anyway**. Follow the wizard and click **Finish**.

### 3.2 Running without installing (portable copy)

Extract `ErrinILS-Windows.zip` to any folder or USB flash drive and double-click `ErrinILS.exe`. It runs portably and keeps data in the same user profile directory.

### 3.3 Where your data is stored

Everything lives in `%APPDATA%\Errin\` (press Win+R and type `%APPDATA%\Errin`):

| File Name | Purpose and Contents |
| --- | --- |
| `errin-data.json` | Main database: catalog, patrons, loans, holds, fines, and policy rules. |
| `errin-data.bak` | Automatic rollback copy: saved before every write. Recovers if sudden power loss occurs. |
| `errin-config.json` | Administrator password hash (PBKDF2/SHA-256), Z39.50 settings, and paired device tokens. |

# 6. The Dashboard & Daily Overview

The Dashboard is the first page you see upon launching ErrinILS. It provides an immediate overview of active operations without requiring any manual data entry.

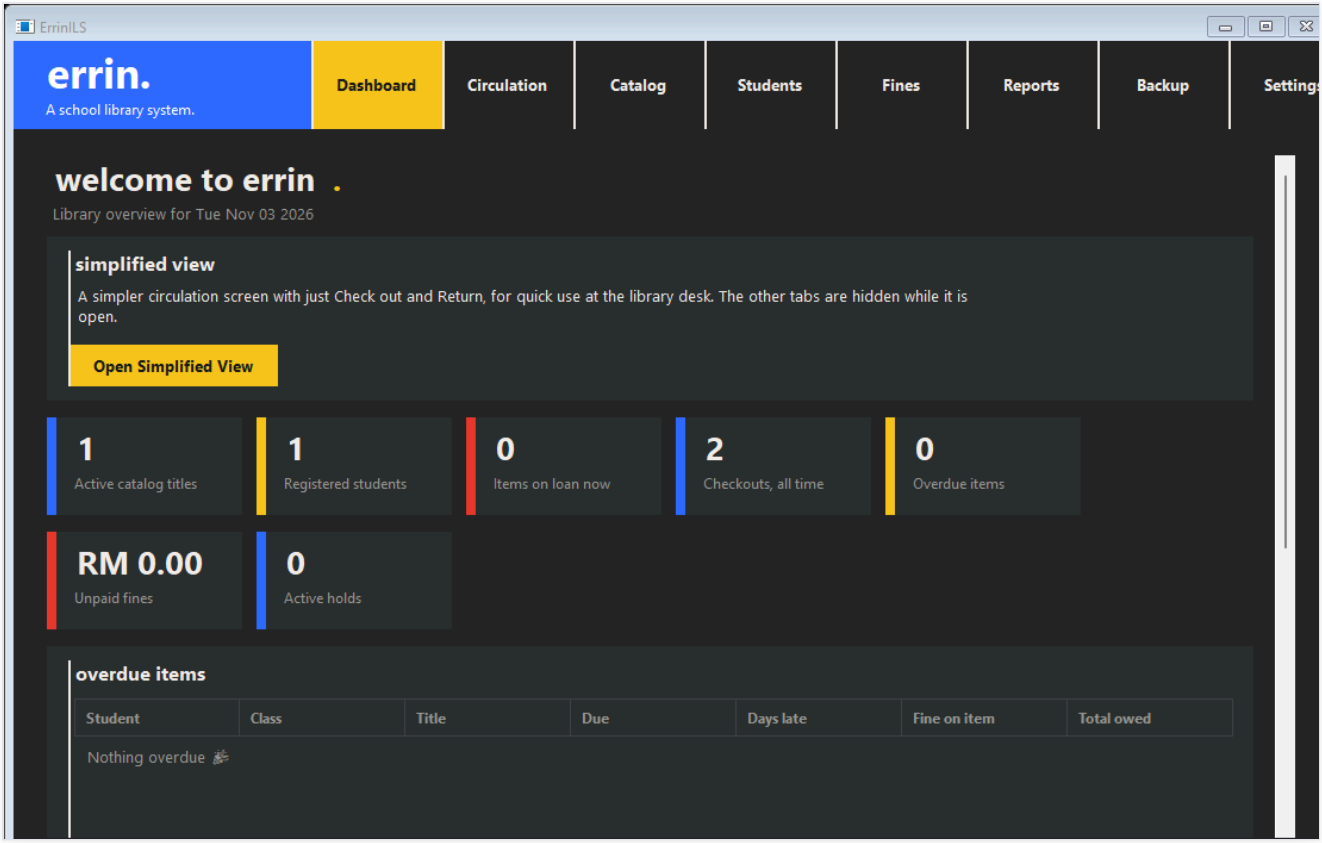



**Figure 1: The ErrinILS Dashboard — Active collection metrics, real-time overdue notices table, and reading leaderboard.**

## 6.1 What the Dashboard shows

- **Simplified View card:** One-click button to launch the student kiosk mode.
- **Metric Counters:** Active catalogue titles, registered students, books out on loan, all-time checkout tally, overdue book count, and total unpaid fines.
- **Overdue Items table:** Lists student name, class, book title, due date, days late, and total monetary balance owed. Use this table every morning to issue reminder slips.
- **Top 10 Borrowers:** Quick ranking of your top readers this month, this year, and of all time.

# 7. Circulation: Check Out, Return, Renew and Holds

Circulation is the busiest desk in the library. Click the **Circulation** tab; the cursor is automatically placed in the active scan box.

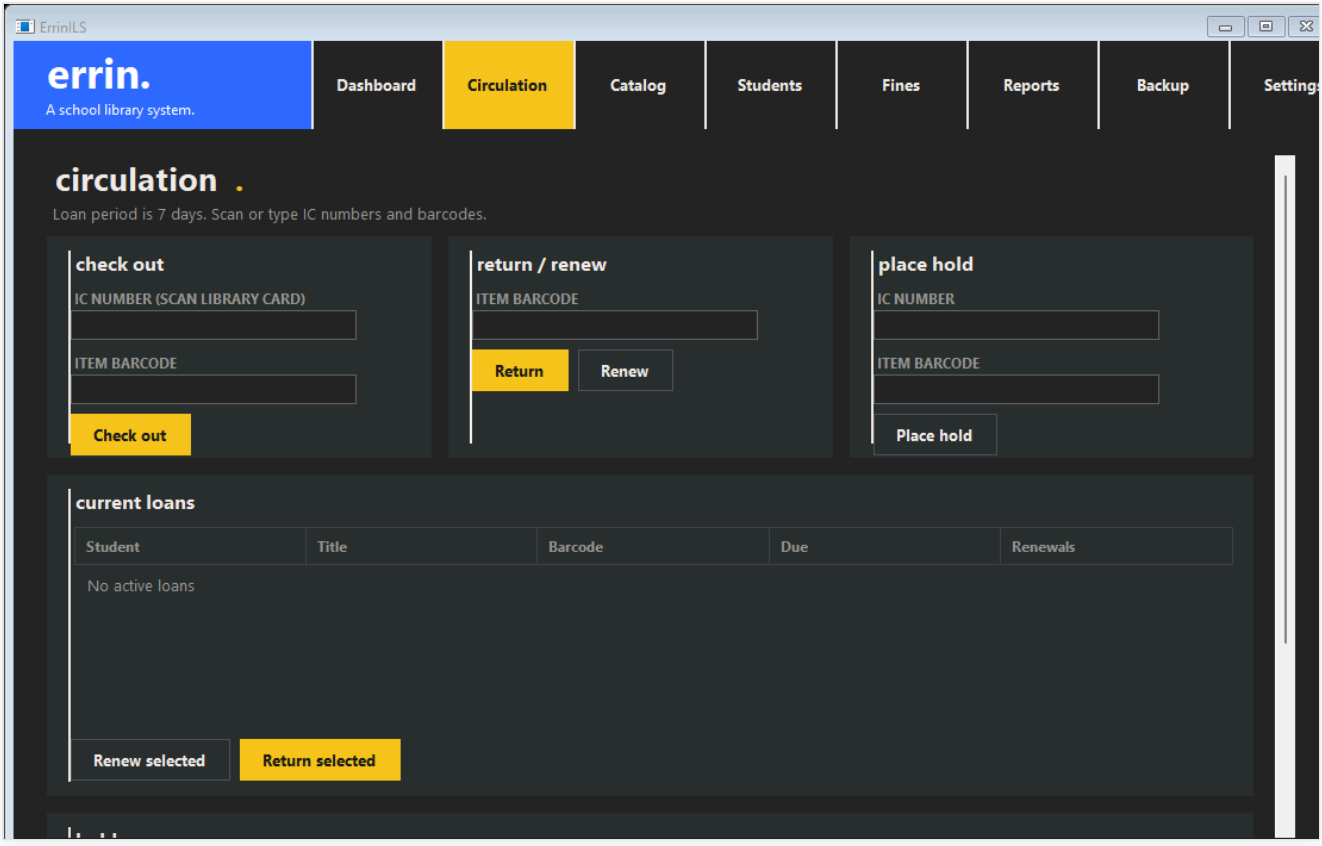



**Figure 2: The Circulation Desk — Fast barcode checkout, returns/renewals box, active loans list, and holds queue.**

## 7.1 Checking out a book

1. Scan the student's library card barcode into **IC number** (or type the 12 digits).
2. The cursor automatically advances to **Item barcode**. Scan the book's barcode.
3. The checkout registers immediately. Toast confirms: *"Checked out. Due [YYYY-MM-DD]"*.

## 7.2 Returning and renewing books

- **Return:** Scan the book barcode in **Return / Renew** and press Enter. If late, the fine is assessed immediately. If another student placed a hold, an alert notifies you to put the book aside.
- **Renew:** Scan the book and click **Renew**. Extends loan by 7 days. Blocked if overdue, renewed twice already, or on hold for another reader.

# 8. The Simplified View (Student Self-Checkout Kiosk)

A calm, high-contrast, large-print desk screen with only Check Out and Return, tailored for young students. Administrative toolbars are hidden.

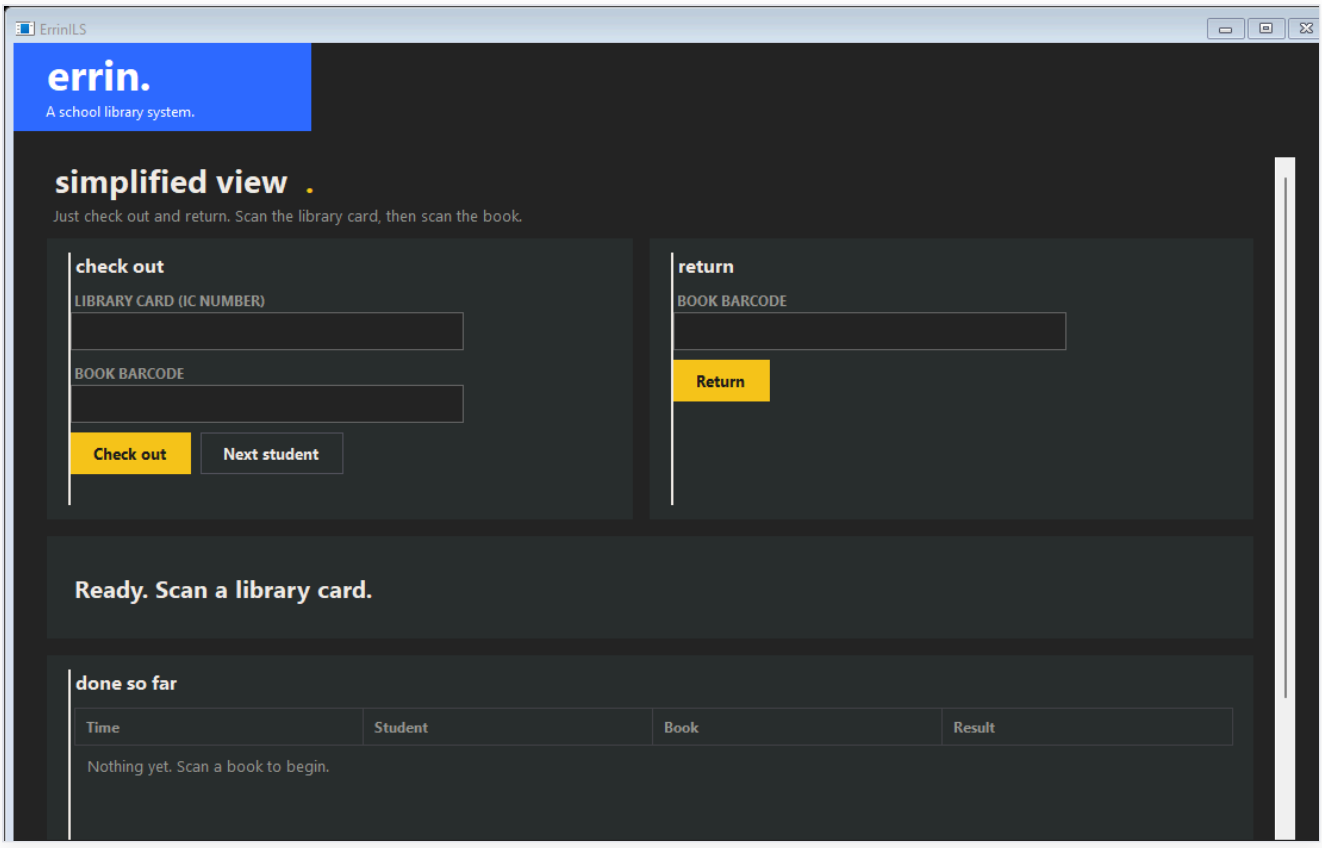



**Figure 3: Simplified Student Kiosk — Child-friendly touch interface, large color banner, and audit transaction log.**

## 8.1 How students use the kiosk

1. The screen prompts: *"Ready. Scan a library card."*
2. Student scans their card. The banner turns green: *"Hello, [Student Name]!"*
3. Student scans the book barcode. The kiosk plays a cheerful chime: *"Checked out: [Title]. Due back [Date]!"*
4. Click **Next student** to clear the screen for the next reader.

> **Exiting Kiosk Mode**
>
> Click **Exit Simplified View** in the bottom right corner. Entering the administrator password is required to return to the full administrative desk.

# 9. The Catalog & 5-Source ISBN Copy-Cataloguing

The Catalog tab manages all bibliographic records and physical copies.

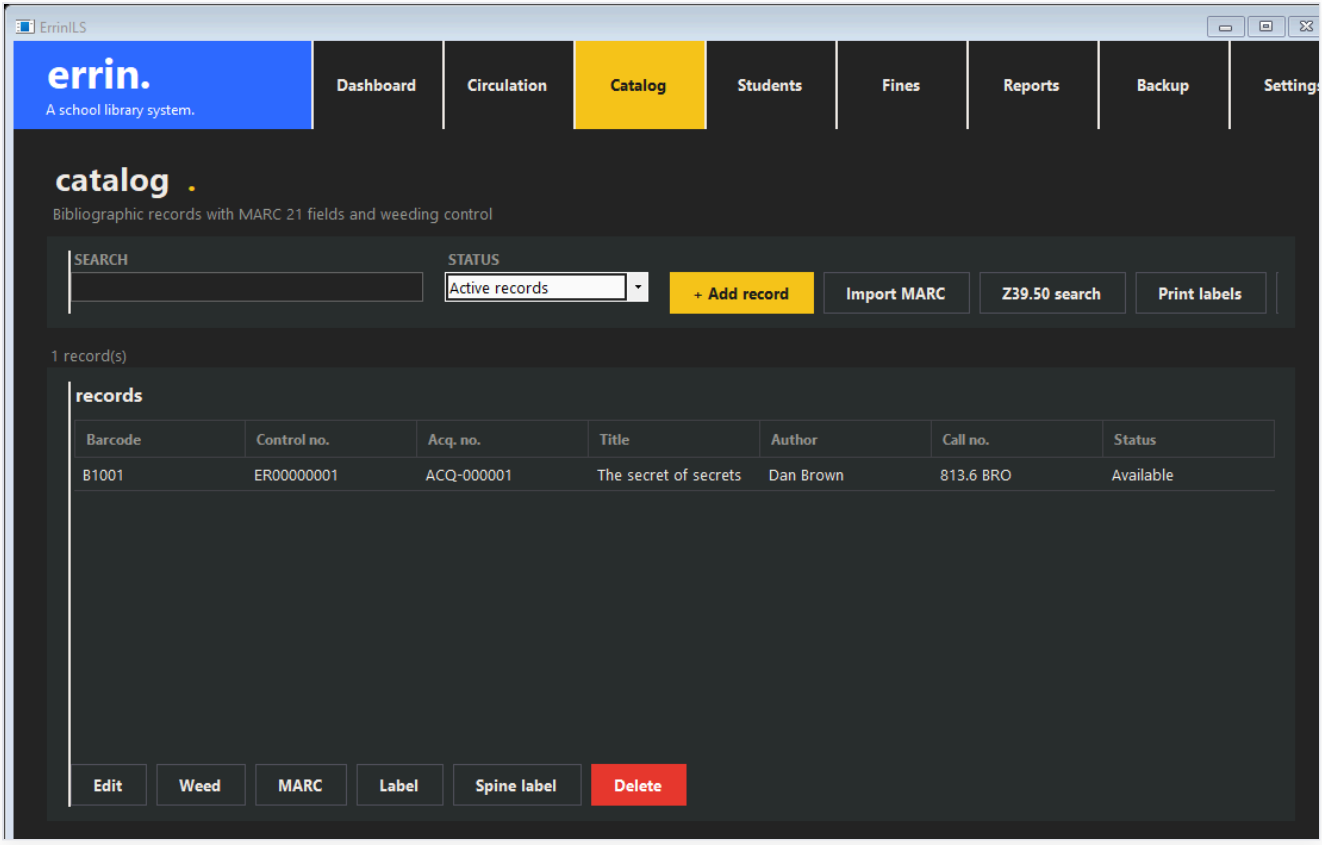



**Figure 4: The Bibliographic Catalog — Multi-term searching, active vs weeded status filtering, and barcode generation.**

## 9.1 5-Source ISBN Copy-Cataloguing

To add a book without manual typing, click **+ Add record**, scan the ISBN in **Copy cataloguing**, and click **Look up**. ErrinILS queries 5 international sources simultaneously:

1. **Library of Congress (Z39.50)**: Direct ASN.1 BER connection (`lx2.loc.gov:210`).
2. **Open Library**: REST API bibliographic repository.
3. **Google Books API**: Commercial book metadata, summaries, and publication years.
4. **HathiTrust**: Academic union catalogue via MARCXML.
5. **K10plus German Union**: SRU MARCXML gateway.

Title, author, publisher, publication year, subjects, and Dewey Decimal classification are auto-populated.

# 10. Students & Patron Management

Manage registered students, track class allocations, and print scannable library cards.

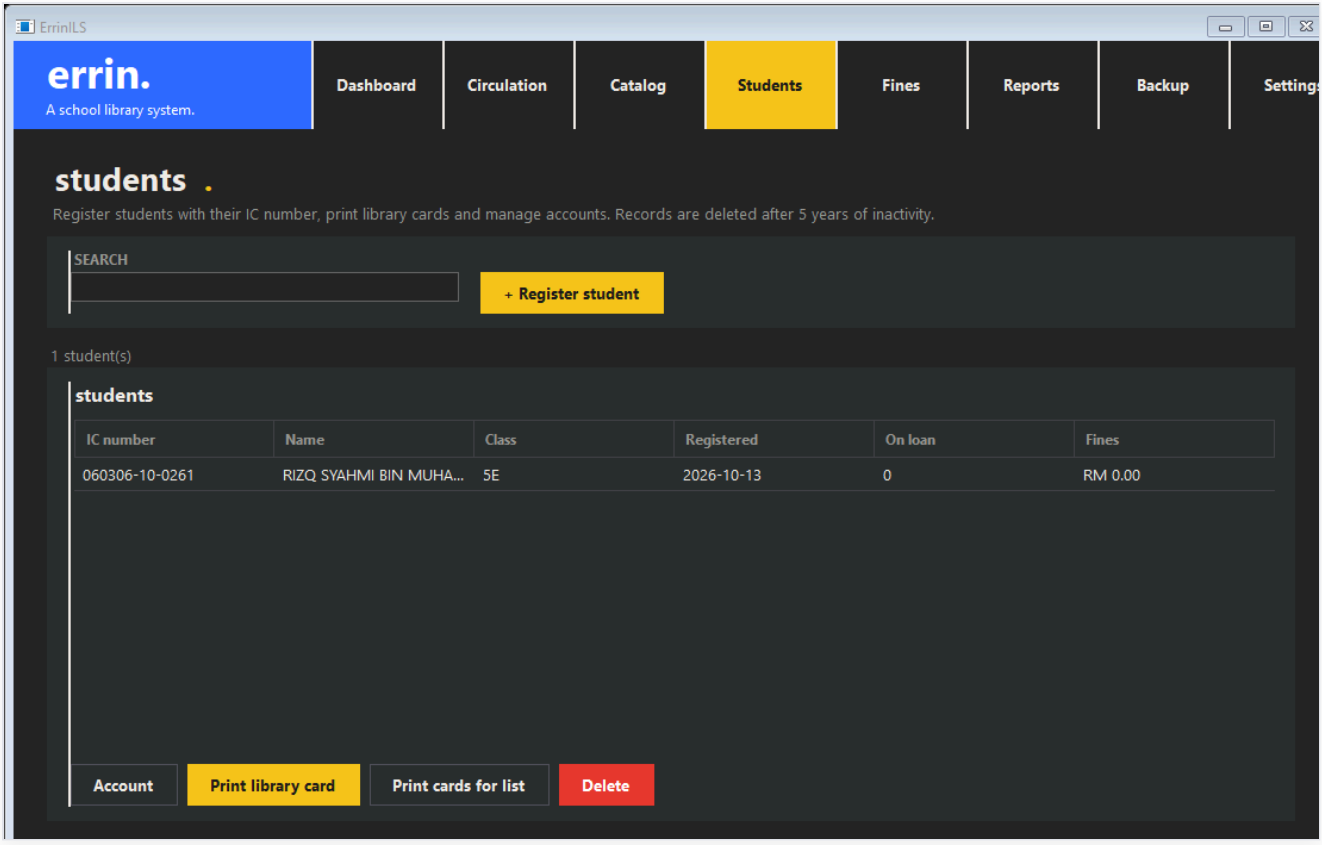



**Figure 5: Student & Patron Management — Malaysian IC verification, borrowing histories, and card printing.**

## 10.1 Registering students & Malaysian IC verification

Click **+ Register student**. Enter the 12-digit Malaysian IC or MyKid number. ErrinILS formats the number into `YYMMDD-PB-###G` and strictly verifies that the birth month and calendar day are mathematically valid. Enter the student's full name, class (e.g. `4 Melur`), and print their scannable card (85.6 × 54 mm).



# 11. Fines Management & Lost Books

Fines are assessed when late books are returned: `Days Overdue × Daily Rate (RM 0.10)`.

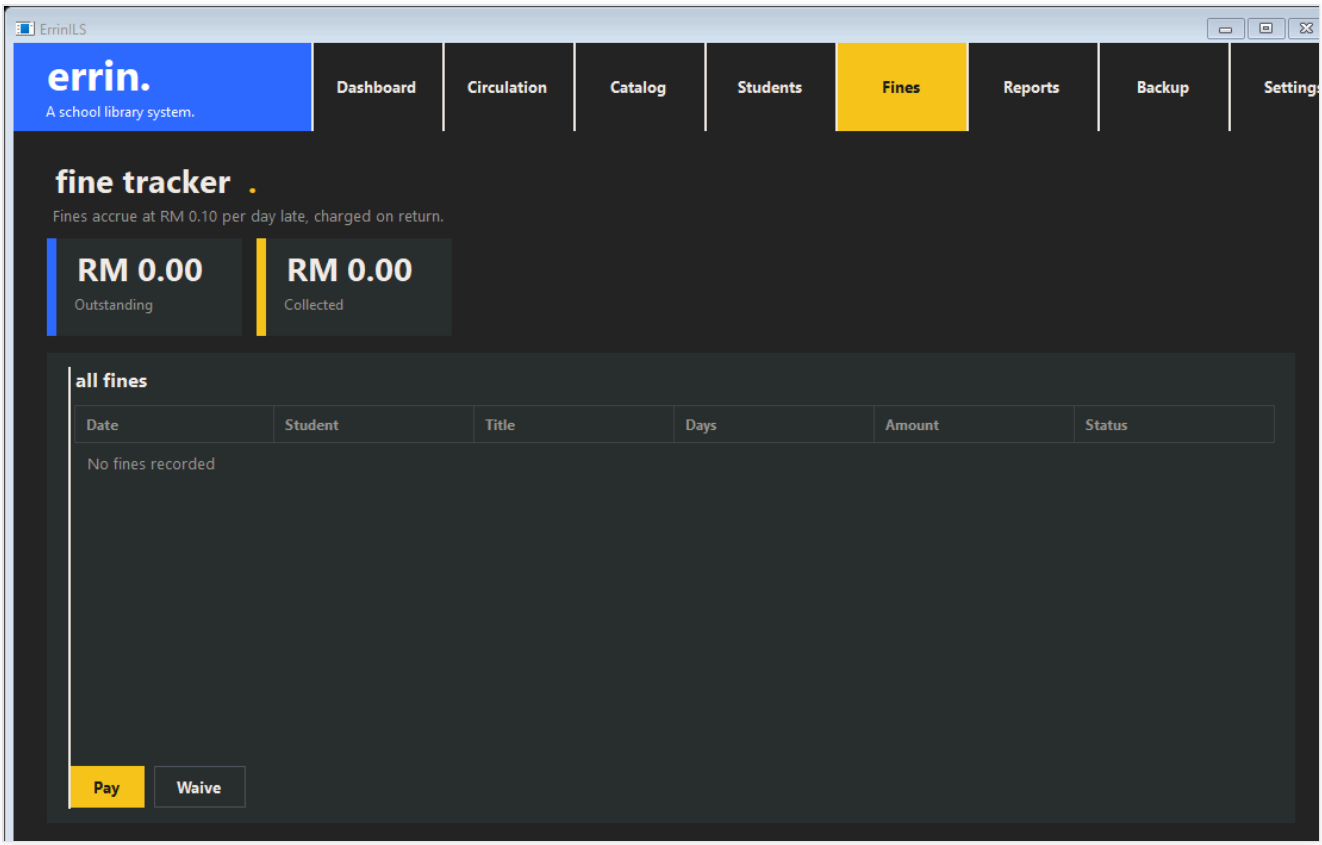



**Figure 6: The Fine Tracker — Real-time outstanding balances, payment recording, and fine waivers.**

Select any fine record in the table and click **Pay** when the student pays cash, or **Waive** if excusing the fee. Settling fines immediately lifts the borrowing block.

# 13. Remote Access, QR Code Pairing & Settings

ErrinILS features a built-in pure C# HTTP server (`0.0.0.0`) enabling wireless access from smartphones and tablets across school Wi-Fi and hotspots.

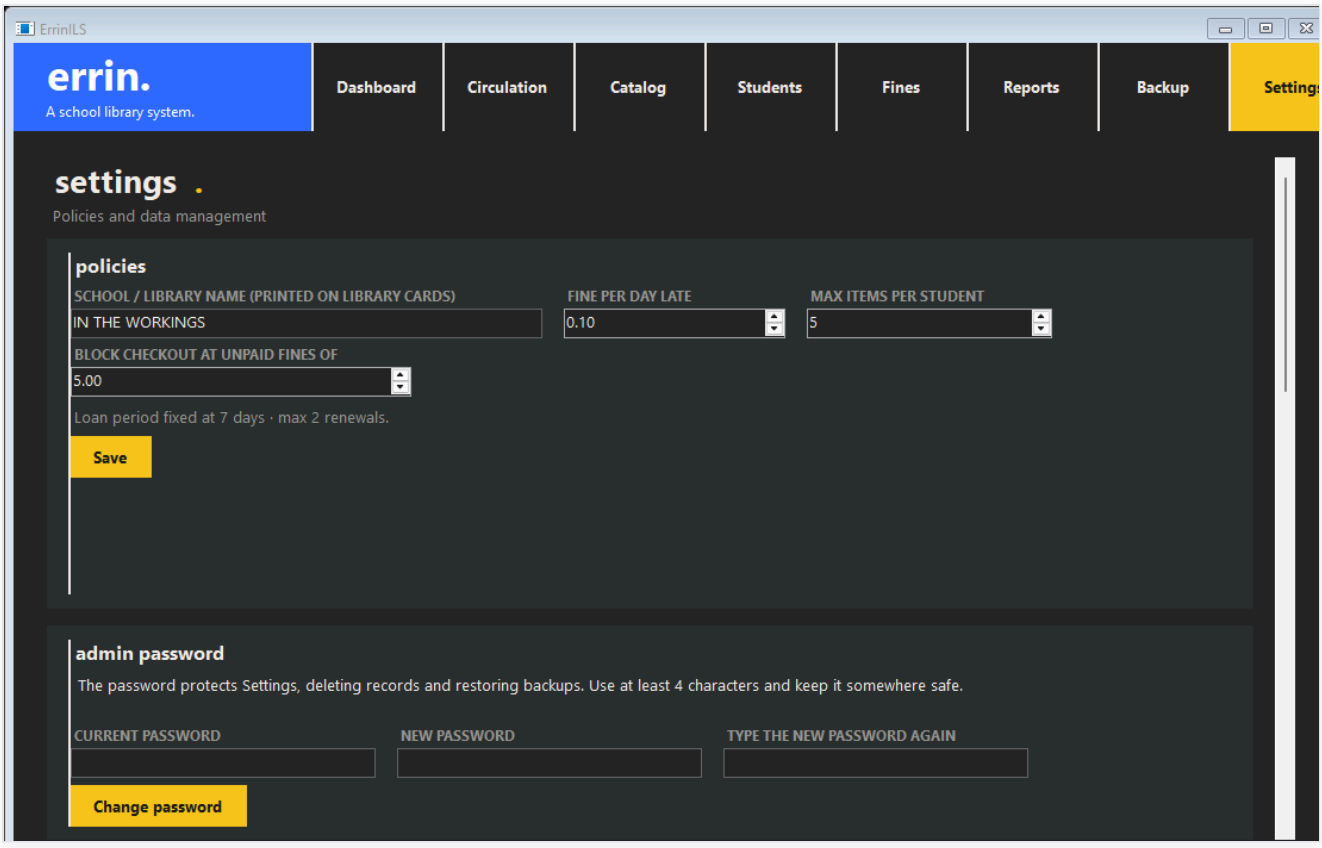



**Figure 8: Settings & Remote Access — Fine rates, administrator password, and wireless QR code device pairing.**

## 13.1 Instant QR Code & PIN pairing

1. In **Settings** → **Remote Access**, click **Pair New Device**.
2. Scan the generated QR code with your smartphone camera.
3. The mobile browser opens the web app with the 6-digit PIN pre-filled. Tap **Verify & Pair**.
4. A persistent 32-byte cryptographic token (`X-Errin-Token`) authenticates the device. Use your phone camera as a wireless shelf scanner.

# 14. Reports, Reading Champions & Collection Analytics

Open the **Reports** tab for executive library intelligence and Nilam reading program tracking.

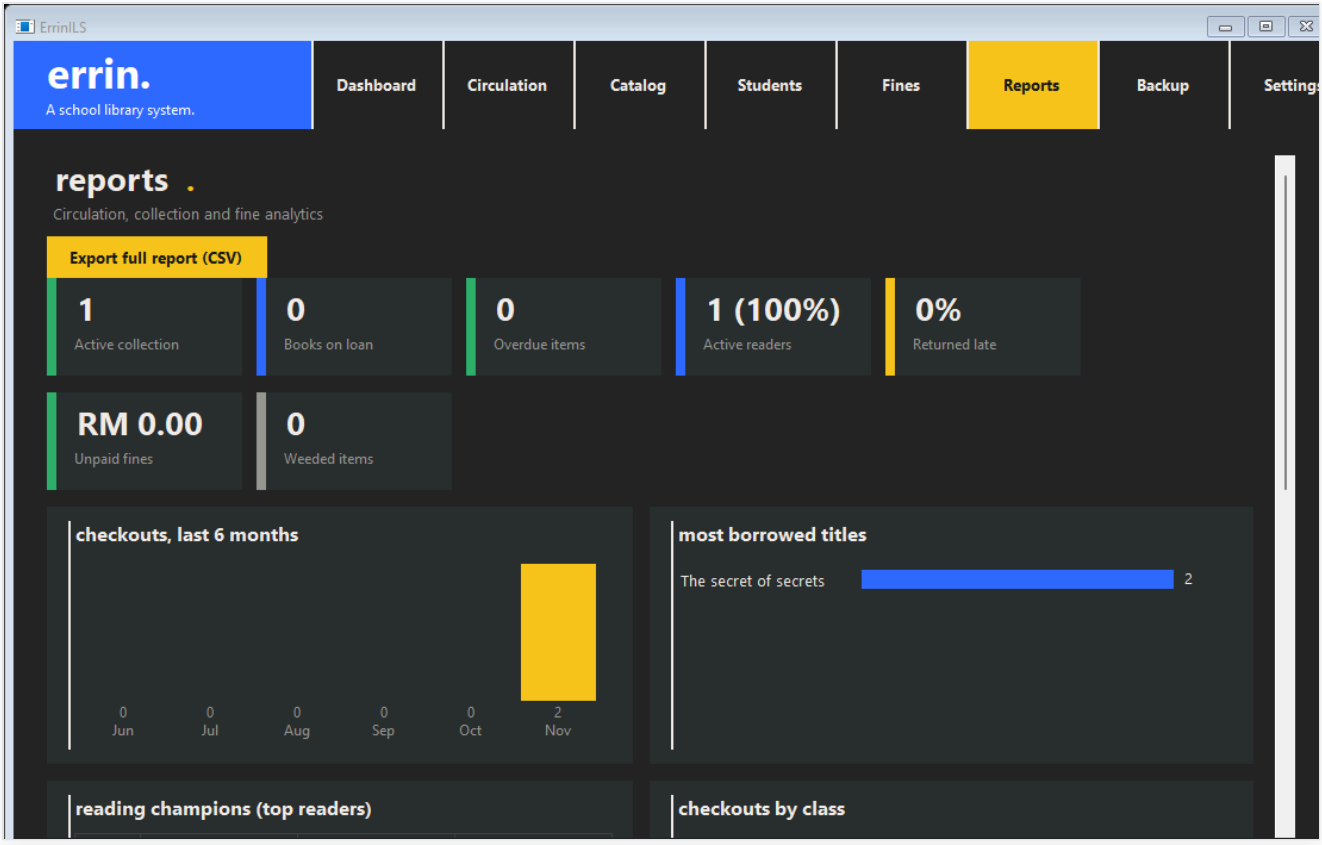



**Figure 7: Reports & Reading Champions — Top 25 reader leaderboard, 6-month borrowing trajectory, and Dewey 100s breakdown.**

- **Reading Champions Leaderboard:** Top 25 student readers ranked by total books borrowed.
- **Circulation Trajectory:** Monthly borrowing trends over the last 6 calendar months.
- **Dewey Decimal Breakdown:** Collection distribution across the 10 core DDC classes.
- **Dormant Inventory Audit:** Identifies titles with zero borrowings over the past 12 months.
- **1-Click CSV Full Report:** Exports a complete spreadsheet ready for school board presentations.

# 15. Backup & Disaster Recovery

In the **Backup** tab, click **Export Backup (ZIP)**. Save the file to an external USB flash drive every Friday afternoon. To restore on a new PC, install ErrinILS, open the Backup tab, and click **Restore from backup**.

# 18. Troubleshooting Guide

| Problem Encountered | Likely Root Cause | Exact Solution |
|---|---|---|
| **"Windows protected your PC"** | Unsigned open-source installer. | Click *More info*, then click *Run anyway*. |
| **Setup reports missing .NET** | Windows 7 without .NET runtime. | Install free Microsoft .NET Framework 4.8. |
| **Status bar red: "NOT saving"** | Disk full or write-protected. | Free disk space; export emergency backup to USB. |
| **Scanner does not submit** | Scanner not sending Enter key. | Configure scanner in USB HID Keyboard mode with Enter suffix. |
| **Phone cannot connect to QR Web** | Phone on different Wi-Fi network. | Connect phone to the same Wi-Fi SSID as the PC. |

**Forgot Administrator Password?**

Close ErrinILS, press `Win + R`, enter `%APPDATA%\Errin`, and rename or delete `errin-config.json`. Launch ErrinILS; the password resets to default ( `admin` or `errin` ). Set a new password immediately in Settings.

# Appendix A. Quick Task Reference Guide

| Goal | Tab | Action |
|---|---|---|
| **Lend a book** | Circulation | Scan student card, scan book barcode. |
| **Return a book** | Circulation | Scan book barcode in Return box. |
| **Extend loan duration** | Circulation | Scan book barcode, click Renew. |
| **Add book via ISBN** | Catalog | Click + Add record, scan ISBN, click Look up, Save. |
| **Weed damaged book** | Catalog | Select book, click Weed Copy, choose reason, confirm. |
| **Register student** | Students | Click + Register student, enter 12-digit IC, name, class. |
| **Record fine payment** | Fines | Select fine record, click Pay. |
| **Pair phone via QR** | Settings | Click Pair New Device, scan QR with camera. |
| **Export backup to USB** | Backup | Click Export Backup (ZIP), save to flash drive. |

# Appendix B. File and Directory Reference

| Component | Standard File System Location |
|---|---|
| **Program Binary** | `C:\Program Files (x86)\ErrinILS\ErrinILS.exe` |
| **Main Database** | `%APPDATA%\Errin\errin-data.json` |
| **Automatic Rollback File** | `%APPDATA%\Errin\errin-data.bak` |
| **Config & Auth Tokens** | `%APPDATA%\Errin\errin-config.json` |
| **Online Documentation** | https://rzsy.github.io/ErrinILS/guide.html |
| **Official Repository** | https://github.com/RZSY/ErrinILS |

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